A MAJOR PROJECT REPORT
(PHASE-I)

ON

DUALNET-AL: BI-NETWORK META-LEARNING FRAMEWORK FOR MULTI-CLASS ACTIVE LEARNING

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By

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CANDIDATE'S DECLARATION

I hereby certify that the work presented in the **Phase-I report** of the project entitled **"DualNet-AL: Bi-Network Meta-Learning Framework for Multi-Class Active Learning"**, in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computer Science and Engineering, IILM University, Greater Noida**, is an original work carried out by me under the supervision of **Dr. Jaswinder Singh**, School of Computer Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by me for the award of any other degree or qualification at this or any other institution.

Lakshya Sharma (2410031752) (3CSE9)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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ABSTRACT

Deep learning models depend on large volumes of labelled data, and in most real-world multi-class settings annotation remains the main bottleneck to building a usable classifier. Active learning reduces this cost by letting a model select, from a large pool of unlabelled data, only the instances most worth labelling next. Most existing query strategies, however, rank instances using a single hand-designed signal such as predictive uncertainty, predicted loss, or feature-space diversity, and were not designed with class imbalance in mind. Because minority classes generate few confident signals early in training, a heuristic tuned around overall uncertainty tends to keep favouring already well-represented classes, leaving rare categories under-sampled across the labelling budget.

This project proposes **DualNet-AL**, a bi-network meta-learning framework for multi-class active learning. The framework separates the classification objective from the sample-selection objective through two cooperating networks: a **Task Network**, a standard CNN classifier trained on the currently labelled pool, and a **Query Network**, a lightweight meta-learner trained to score every unlabelled instance by its expected labelling benefit. The Query Network is trained episodically across successive labelling rounds so that its scoring behaviour is learned from data and generalises across datasets rather than being fixed by a designer. A class-balance signal, computed from the current label distribution, is supplied to the Query Network so that its selections work to correct existing imbalance rather than reinforce it.

The framework will be evaluated through a pool-based active learning simulation on public benchmark datasets, including CIFAR-10, CIFAR-100, and a long-tailed CIFAR variant, and compared against established baselines such as random sampling, entropy-based uncertainty sampling, Core-Set selection, and Learning Loss. Performance will be measured through accuracy-versus-budget learning curves, per-class F1-score, and the area under the learning curve. Phase I of the project defines the problem, surveys the relevant literature, states the objectives and boundaries of the work, and evaluates the novelty and feasibility of the proposed system before implementation begins in Phase II.

1. PROBLEM DEFINITION

Modern deep neural networks achieve strong classification performance, but that performance is closely tied to the volume and quality of labelled training data. In most practical settings, unlabelled data is abundant while labelled data is scarce, because annotation demands expert time and effort. This cost grows further as the number of target classes increases, since a multi-class problem needs annotated examples spread meaningfully across every class rather than simply a large total count.

Active learning addresses this bottleneck by letting the model itself choose, from a large pool of unlabelled data, which instances would be most useful to label next, and then requesting labels only for that subset. Widely used query strategies rank instances using a single hand-designed signal, such as model uncertainty, predicted loss, or diversity in feature space. Two limitations follow from this in multi-class settings. First, a heuristic that works well on one dataset or architecture does not necessarily transfer its effectiveness to a different one, since “informativeness” is itself data-dependent. Second, most query strategies are class-agnostic: they rank samples by an aggregate confidence or diversity score without considering how the resulting selections are distributed across classes, so classes that are already well represented keep attracting most of the new labels while rare classes stay under-sampled, and model performance on those rare classes remains weak as a result.

This project addresses that gap through a bi-network meta-learning framework that learns, directly from data, how to score unlabelled instances for annotation while explicitly accounting for the current class distribution of the labelled pool. The system is intended as a labelling-cost-reduction tool for multi-class

classification pipelines, and not as a replacement for the annotators or domain experts who ultimately assign the labels.

2. LITERATURE SURVEY

The review examined work published mainly between 2018 and 2025 on active learning for deep classifiers, covering uncertainty- and loss-based query strategies, meta-learned and transferable query policies, and methods that explicitly address class imbalance during sample selection. Table 1 summarises the studies used to shape the proposed approach.

S.No.	Author(s) & Year	Technique Used	Key Findings	Limitations
1	Sener & Savarese (2018)	Core-Set selection in CNN feature space	Formulates active learning as a coreset-selection problem with a theoretical bound linking the labelling gap to the coverage radius of selected points.	Relies on Euclidean distance in a high-dimensional feature space, which weakens as dimensionality grows; selections are not class-balance aware.
2	Yoo & Kweon (2019)	Learning Loss for Active Learning	Attaches a small, task-agnostic loss-prediction module to the target network to rank unlabelled samples by predicted loss.	Samples with the largest predicted loss can be outliers or noisy labels; no explicit handling of class imbalance.
3	Pang, Dong, Wu & Hospedales (2018)	Meta-learned query policy via deep reinforcement learning	Treats the active-learning query criterion itself as a meta-learning problem, training a deep network to predict the best point to annotate so that the policy transfers across datasets.	Evaluated only with a binary base classifier; extending the state representation to deep multi-class classifiers remains open.
4	Ren, Xiao, Chang *et al.* (2021)	Survey and taxonomy of deep active learning	Organises deep active-learning methods by annotation type, query strategy, model architecture, and training process; notes that batch-mode, uncertainty-based, and hybrid strategies dominate current practice.	Reported gains vary widely across benchmarks and are rarely compared under identical experimental protocols.
5	Zhang, Shao, Verma & Nowak (2023)	TAILOR – Thompson-sampling algorithm selection for imbalanced multi-class/multi-label active learning	Adaptively selects among several existing query strategies using reward functions aimed at gathering class-balanced examples, matching or beating the best individual strategy.	Chooses among fixed, pre-existing heuristics rather than learning a new per-sample scoring function end to end.
6	Nuggehalli, Zhang, Jain & Nowak (2024/2025)	DIRECT – active learning under class imbalance and label noise	Locates class-separation boundaries and queries the most uncertain nearby examples, reporting substantial annotation-cost savings on imbalanced benchmarks.	Threshold-based separation is designed for the imbalance-plus-noise setting and is not framed as a general, transferable meta-learned policy.

2.1 Research Gaps Identified

- Most class-balance-aware active-learning methods (TAILOR, DIRECT, and similar class-balanced approaches) select among existing fixed heuristics rather than jointly training a transferable per-sample scoring network.
- Meta-learned query policies have mainly been validated on binary classifiers; their behaviour with deep multi-class classifiers and an explicit class-balance signal is comparatively unexplored.
- Reported gains across active-learning papers are rarely compared under one consistent multi-class benchmark protocol, which makes it hard to judge whether a new strategy actually generalises.
- Few frameworks combine a learned meta-scoring network with an explicit, per-round class-distribution signal inside a single end-to-end trainable system.

3. OBJECTIVES

- Design a bi-network architecture consisting of a Task Network (CNN classifier) and a Query Network (meta-learner) that interact across successive active-learning rounds.
- Train the Query Network to score unlabelled instances using signals from the Task Network, such as predictive uncertainty and feature embeddings, combined with a class-balance signal drawn from the current labelled pool.
- Implement a pool-based, batch active-learning simulation loop on public multi-class image benchmarks.
- Compare DualNet-AL against random sampling, entropy-based uncertainty sampling, Core-Set selection, and Learning Loss as baseline query strategies.
- Evaluate results using accuracy-versus-budget learning curves, per-class F1-score, area under the learning curve, and the reduction in class-imbalance ratio across rounds.
- Provide a lightweight dashboard to visualise the learning curves and the class distribution of the labelled pool at each active-learning round.

4. SCOPE

4.1 In Scope

- Multi-class image classification on standard benchmark datasets (CIFAR-10, CIFAR-100, and a long-tailed CIFAR variant) under a simulated oracle, where ground-truth labels are revealed only for the instances chosen for annotation.
- A pool-based, batch-mode active-learning loop implemented and run offline.
- The bi-network design (Task Network + Query Network), evaluated against four established baseline query strategies.
- A lightweight dashboard for visualising learning curves and label-distribution results.

4.2 Out of Scope

- Deployment with real human annotators in the loop; labelling is simulated using existing dataset ground truth.
- Non-image modalities such as text or tabular data, which are noted as a possible future extension.
- Streaming or online active learning; only the pool-based batch setting is addressed in this project.

- Integration with a production data-labelling platform as a finished, deployable plug-in.

5. NOVELTY

Active learning and meta-learning are both established research areas individually, so this project does not claim novelty from either idea alone. Its contribution lies in combining three practical choices within one student-scale prototype: a learned and transferable scoring function, explicit class-balance conditioning, and a bi-network separation of concerns.

- **Learned, transferable scoring instead of a fixed heuristic:** the Query Network is trained across active-learning rounds so that its selection behaviour is learned from data rather than fixed in advance by a designer, addressing the transfer limitation seen in hand-crafted strategies.

- **Explicit class-balance conditioning:** an imbalance signal computed from the current labelled pool is supplied as an input to the Query Network, so that class balance is optimised jointly with informativeness inside a single scoring function rather than being layered on afterward.

- **Bi-network separation of concerns:** keeping the Task Network and Query Network as separate, cooperating modules lets the classifier focus purely on classification while the meta-learner focuses purely on sample valuation, making it possible to swap either component independently in later work.

6. FEASIBILITY STUDY AND ANALYSIS

Dimension	Assessment
Technical Feasibility	PyTorch and torchvision are mature, well documented, and freely available. CIFAR-family datasets are small enough to iterate quickly on free-tier GPUs, and episodic meta-training loops are well documented in existing frameworks. The team already has working knowledge of Python and introductory machine learning.
Economic Feasibility	Python, open-source machine-learning libraries, public benchmark datasets, and free Google Colab or Kaggle notebook tiers cover the main requirements. No licence fee is expected. Optional paid GPU credits may be used only if larger-scale runs on CIFAR-100 or a long-tailed variant require longer training time.
Operational Feasibility	A simulation-based workflow, where the oracle is the dataset's own ground-truth label, keeps the operational loop simple to run and repeat, since no live annotators are required for the Phase II experiments.
Ethical Feasibility	The project uses only public, non-personal image benchmark datasets, so no patient or personally identifiable data is involved. Correcting class imbalance is itself an ethical motivation for the work, and every result will be presented as research evidence for reducing labelling cost, not as a deployed decision-making system.

7. MULTIDISCIPLINARY AND TRANSDISCIPLINARY ASPECTS

The project draws on methods from computing, mathematics, data science, human-computer interaction, and responsible-AI practice. Each discipline contributes to a different part of the problem, as summarised below.

Discipline	Contribution to the Project
Computer Science & Engineering	Design of the Task Network and Query Network, algorithm implementation, software development, testing, and integration of the active-learning loop.

Discipline	Contribution to the Project
Mathematics & Statistics	Loss functions, meta-optimisation across labelling rounds, probability-based classification, and the statistical evaluation measures used to compare strategies.
Data Science	Dataset curation, preprocessing, augmentation, exploratory analysis, and the construction of the long-tailed benchmark used to study class imbalance.
Human-Computer Interaction (HCI)	Design of a clear dashboard for visualising learning curves, per-class performance, and the evolving label distribution across active-learning rounds.
Ethics & Responsible AI	Fairness considerations in class-balanced sample selection, transparent reporting of limitations, and treatment of the system as a research prototype rather than a production labelling tool.

8. PRODUCT DEVELOPMENT SCOPE

Although the present outcome is an academic prototype, the active-learning pipeline provides a basis for later product development. The immediate deliverable and possible extensions are separated below so that future ideas do not expand the approved project boundary.

8.1 Minimum Viable Product (MVP) - Target of this Project

- Run a simulated, pool-based active-learning loop using the DualNet-AL bi-network on at least two benchmark datasets.
- Produce comparison plots of DualNet-AL against the four baseline query strategies.
- Provide a dashboard showing per-round accuracy, per-class F1-score, and the evolving label distribution.
- Allow the demonstration results to be downloaded or printed for review.

8.2 Potential Future Product Extensions (Beyond Current Scope)

- Connect the framework to a real annotator-in-the-loop labelling interface, such as Label Studio.
- Extend the bi-network approach to text classification and tabular data.
- Extend the approach to object detection and segmentation active learning.
- Package the framework as a plug-in module for existing MLOps or data-labelling pipelines.

9. IDENTIFICATION OF COMPONENTS AND BILL OF MATERIALS (BOM)

This is primarily a software project. Its bill of materials therefore lists programming tools, libraries, datasets, computing services, storage, and existing hardware rather than manufactured components. Estimated costs remain approximate.

S.No.	Component / Resource	Type	Purpose / Role in Project	Cost (Approx.)
1	Python 3.x	Software / Language	Primary language for data preparation, model development, and integration.	Free (Open Source)

S.No.	Component / Resource	Type	Purpose / Role in Project	Cost (Approx.)
2	PyTorch & torchvision	Software / Framework	Construction, training, and evaluation of the Task Network and Query Network.	Free (Open Source)
3	NumPy, Pandas, scikit-learn	Software / Library	Data handling, baseline query-strategy implementations, and metric computation.	Free (Open Source)
4	Matplotlib / Seaborn	Software / Library	Plots for learning curves, per-class F1-score, and label-distribution charts.	Free (Open Source)
5	CIFAR-10 / CIFAR-100 / Long-Tailed CIFAR	Dataset	Labelled multi-class image data for simulated active-learning experiments.	Free (Public Dataset)
6	Google Colab / Kaggle Notebooks (GPU)	Compute Resource	GPU-supported environment for model training and active-learning simulation.	Free tier / Optional paid credits
7	Streamlit / Flask	Software / Framework	Dashboard for visualising active-learning rounds and comparison results.	Free (Open Source)
8	Git & GitHub	Version Control	Source-code management, collaboration, and version history.	Free
9	Local workstation / laptop (minimum 8 GB RAM)	Hardware	Local development, testing, and execution of the dashboard.	Existing (Self-owned)
10	Cloud storage (Google Drive)	Storage	Storage for datasets, model checkpoints, experiment results, and documents.	Free tier

The estimated cost across Phases I–III is negligible because the central tools and datasets are open source, publicly available, or covered by free academic tiers. Extra expenditure would arise only if longer or larger-scale training runs require paid cloud GPU time.

10. CONCLUSION OF PHASE I

Phase I has established a clear foundation for DualNet-AL. It defined the labelling-cost problem in multi-class active learning, reviewed the relevant literature on query strategies, meta-learned policies, and class-balance-aware selection, and converted the intended outcome into measurable objectives. The review also separated the approved Phase I–III deliverable from later product ideas and identified the project's intended contribution: a bi-network framework that pairs a learned, transferable Query Network with explicit class-balance conditioning for multi-class active learning.

The technical, economic, operational, and ethical assessments indicate that implementation is achievable within the academic setting. Required software is available without licence cost, suitable public benchmark datasets exist, and the candidate can begin with existing Python and machine-learning knowledge. The multidisciplinary analysis and bill of materials further clarify the expertise and resources needed for the next phase. Phase II can therefore proceed to detailed system design, the Query Network's meta-training procedure, implementation planning, scheduling, and risk control.

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